

Mohammad Abid Hasan

Graduate Research Assistant, Texas Tech University (TTU)

has24395@ttu.edu – (512) 667 3468 – Portfolio

Education

Texas Tech University, Lubbock, Texas

Jan 2025 – Present

Master of Science in Atmospheric Science

Relevant Coursework: Wind Science and Modeling, Radar Meteorology, Geophysical Fluid Dynamics, Synoptic and Mesoscale Dynamics, Boundary Layer Meteorology

CGPA: 4.00/4.00

Texas State University, San Marcos, Texas

Sep 2024 – Dec 2024

Master of Science in Civil Engineering

Relevant Coursework: Advanced Reinforced Concrete, Advanced Infrastructure Materials, Statistics

CGPA: 4.00/4.00

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh *Mar 2018 – May 2023*

Bachelor of Science in Civil Engineering

Relevant Coursework: Fluid Mechanics, Hydrology, Irrigation and Flood Management, Solid Mechanics, Structural Dynamics

CGPA: 3.52/4.00

Research Interests

- Wind engineering, computational wind engineering (CWE)
- Building relevant winds (damaging straight-line winds, thunderstorms, tornadoes), data assimilation
- Disaster-resilient built environment
- AI-accelerated computational fluid dynamics (CFD)
- Urban canopy layer dynamics, urban heat island
- Structural health monitoring, non-destructive evaluation, and smart infrastructure systems

Publications

Peer-reviewed journal

- **Mohammad Abid Hasan**, Faiaz Khaled, Franklin T. Lombardo, “Advancing tornado-like vortex (TLV) generation in straight-line wind simulators: An LES study,” *Journal of Wind Engineering and Industrial Aerodynamics*, Volume 271, 2026, 106373, ISSN 0167-6105.
DOI: 10.1016/j.jweia.2026.106373

Peer-reviewed journal (to be submitted)

- Faiaz Khaled, **Mohammad Abid Hasan**, Franklin T. Lombardo, “Tornado-like vortex (TLV) induced loading in straight-line wind simulators.”

Conference abstracts and papers

- Oral presentation at the 2026 AMS Annual Meeting, Houston, TX, on *Generating tornado-like vortices (TLVs) in straight-line simulators using moving louvers with LES*

Peer-reviewed conference abstracts and papers (to be presented)

- **Mohammad Abid Hasan**, Faiaz Khaled, Franklin T. Lombardo (2026), “Advancing Tornado-Like Vortices (TLVs) Modeling in Straight-Line Simulators,” *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada
- **Mohammad Abid Hasan**, Faiaz Khaled, Samuel Hanson, Jon Schroeder (2026), “Digital twin of a supercell rear-flank downdraft (RFD) current in a computational domain using LES,” *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada

- Samuel Hanson, **Mohammad Abid Hasan**, Faiaz Khaled (2026), “Investigating interference effects on aerodynamic loading under tornado-like vortices: An LES study,” *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada
- Guangzhao Chen, Faiaz Khaled, **Mohammad Abid Hasan** (2026), “AI/ML-Powered processing of near-surface anemometer network observation for enhancing CFD simulations of low-rise buildings under downburst events,” *The 8th International Symposium on Computational Wind Engineering*, London, ON, Canada

Research Experience

Graduate Research Assistant, Department of Geoscience, Texas Tech University *Jan 2025 – Present*

- Advancing vortex generation in straight-line wind tunnels using Large Eddy Simulation (LES)
- Interaction of transient winds with suburban neighborhoods using LES informed by field data
- Digital twin of a supercell rear-flank downdraft (RFD) current in a computational domain using LES
- Integration of Weather Research and Forecasting (WRF), FastEddy simulation and building-scale CFD to develop a multiscale model

Graduate Research Assistant, Ingram School of Engineering, Texas State University *Sep 2024 – Dec 2024*

- Developed an Artificial Neural Network (ANN) model for structural health monitoring using intrinsic self-sensing concrete data
- Simulated and validated concrete behavior for smart infrastructure applications

Undergraduate Research Thesis, BUET, Dhaka, Bangladesh *2018 – 2023*

- Developed a damage detection model using finite element method (FEM), time series analysis, and ARIMA modeling
- Identified damage locations and severity in simulated bridge models
- Finite Element Modeling, ARIMA, statistical analysis, structural health monitoring

Teaching Experience

Mentored an undergraduate student to develop a machine learning model (XGBoost) to accelerate numerical simulation *Spring 2026*

Lecturer, Department of Civil Engineering, Dhaka International University *Aug 2023 – Jul 2024*

- Taught civil engineering courses, supervised student projects, and guided thesis research
- Developed course materials focused on structural engineering and fluid mechanics.

Project Experience

Real-Time Structural Health Monitoring for Kyle Fire Station *Apr 2023 – Nov 2024*

- Implemented a self-sensing concrete system with vibrating wire strain gauges
- Analyzed real-time data for infrastructure performance using Loggernet

Development of a Swimming Pool Complex, BUET *May 2022 – May 2023*

- Designed and analyzed an Olympic-sized pool using ETABS, AutoCAD, and SketchUp
- Conducted feasibility analysis considering environmental, geological, and economic factors
- Estimated costs and prepared a comprehensive Bill of Quantity (BOQ)

Field Deployment

Obtaining “full-scale” pressure, velocity, and loading data from the field at high spatial and temporal resolution for ABL in Lubbock, Texas *25 Jun 2025 – 27 Jun 2025*

- Deployed TTU StickNets to gather wind data at high spatial resolution

- Gathered high-resolution spatial and temporal wind velocity and pressure data around an instrumented cube

“Full-scale” pressure, velocity, and loading data from the field at high spatial and temporal resolution for vortex based flow system (dust devil) in Las Cruces, New Mexico *29 Jun 2025 – 3 Jul 2025*

- Deployed TTU StickNets and GoPro cameras to visualize and quantify loading on an instrumented cube induced by a dust devil
- Collected near-ground, high-resolution spatial and temporal thunderstorm outflow data

TxDOT 0-7206: Develop Optimum 2-Mat Reinforcement Design in Continuously Reinforced Concrete Pavement (CRCP)

- Conducted finite element simulations to optimize CRCP designs using ANSYS
- Coordinated field data collection and analyzed structural performance using Loggernet software

Skills

- **Programming:** MATLAB, Python, C++ (for numerical simulation)
- **Software:** OpenFOAM, ANSYS, ABAQUS, ETABS, SAP, AutoCAD
- **Hardware:** StickNets, Datalogger (CR1000X), Arduino Mega, vibrating wire strain gauges, steel strain gauges
- **Machine Learning:** Statistical modeling, neural networks (e.g., U-Net)
- Public speaking, technical writing

Test Scores

- **GRE General Test:** 317/340 (Quantitative: 165, Verbal: 152, Analytical Writing: 3.0)
- **TOEFL iBT:** 103/120 (Reading: 25, Listening: 28, Speaking: 25, Writing: 25)

Skills

- **Programming:** MATLAB, Python, C++ (for numerical simulation), high-performance computing
- **Software:** OpenFOAM, ANSYS, ABAQUS, ETABS, SAP, AutoCAD
- **Hardware:** StickNets, Datalogger (CR1000X), Arduino Mega, vibrating wire strain gauges, steel strain gauges
- **Machine Learning:** Statistical modeling, neural networks (e.g., U-Net)
- Public speaking, technical writing